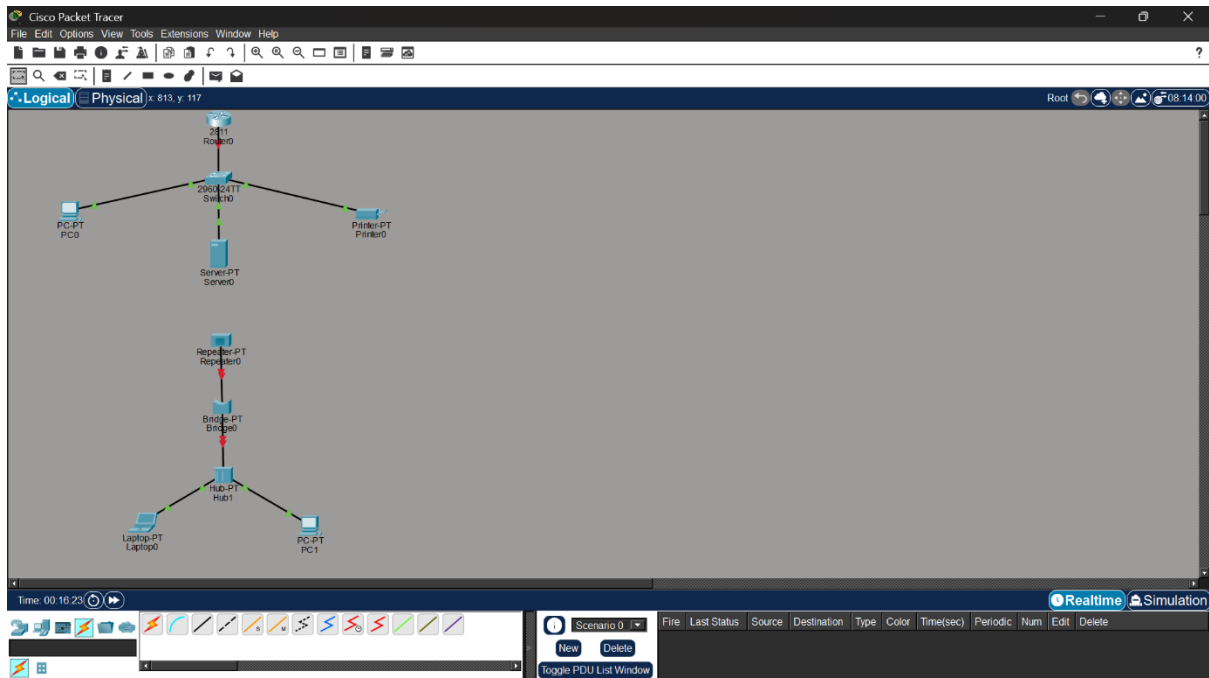
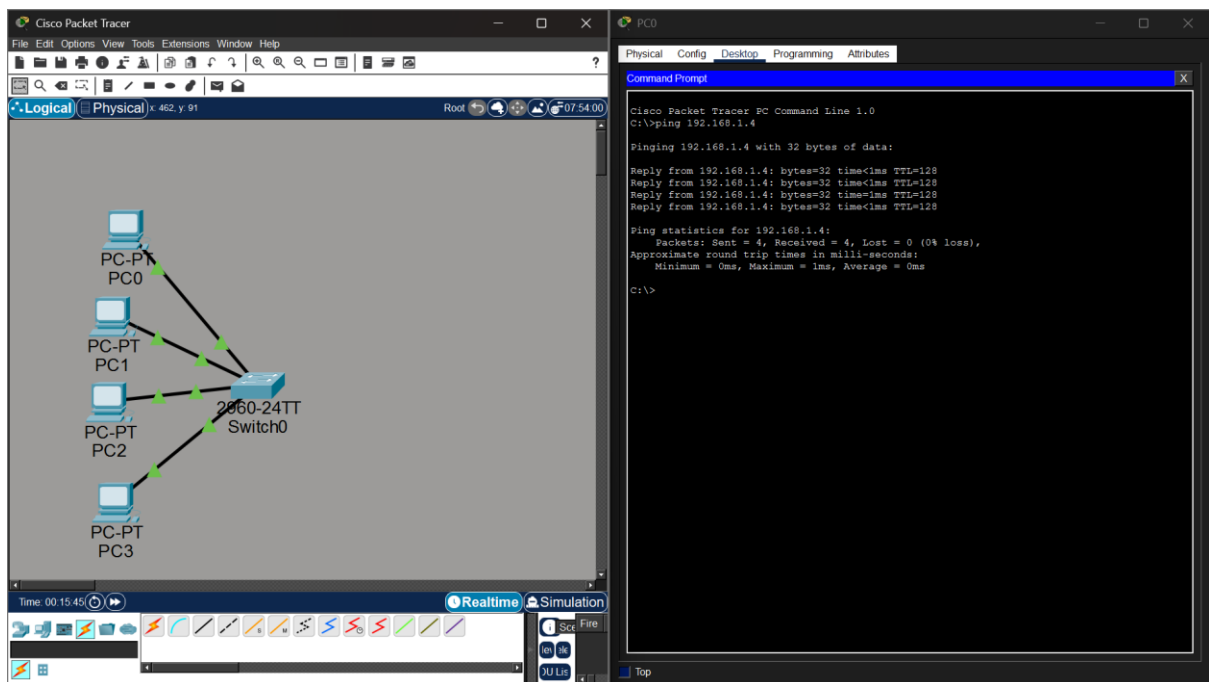


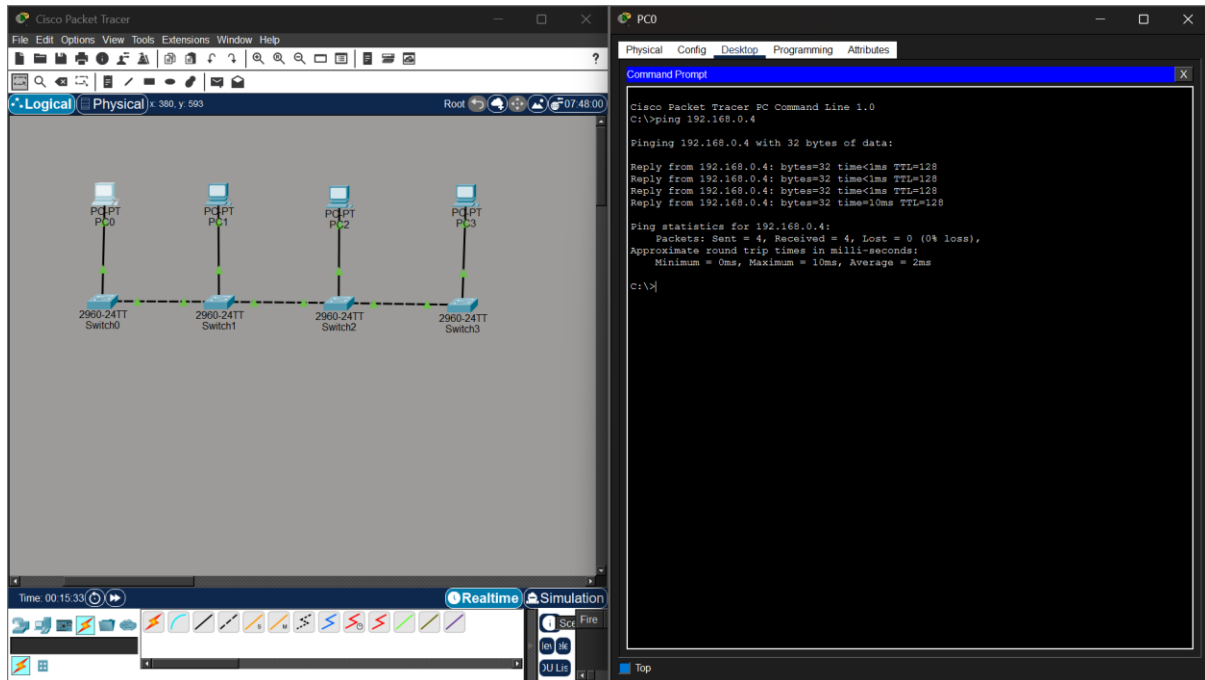
Experiment-01-Network-Components



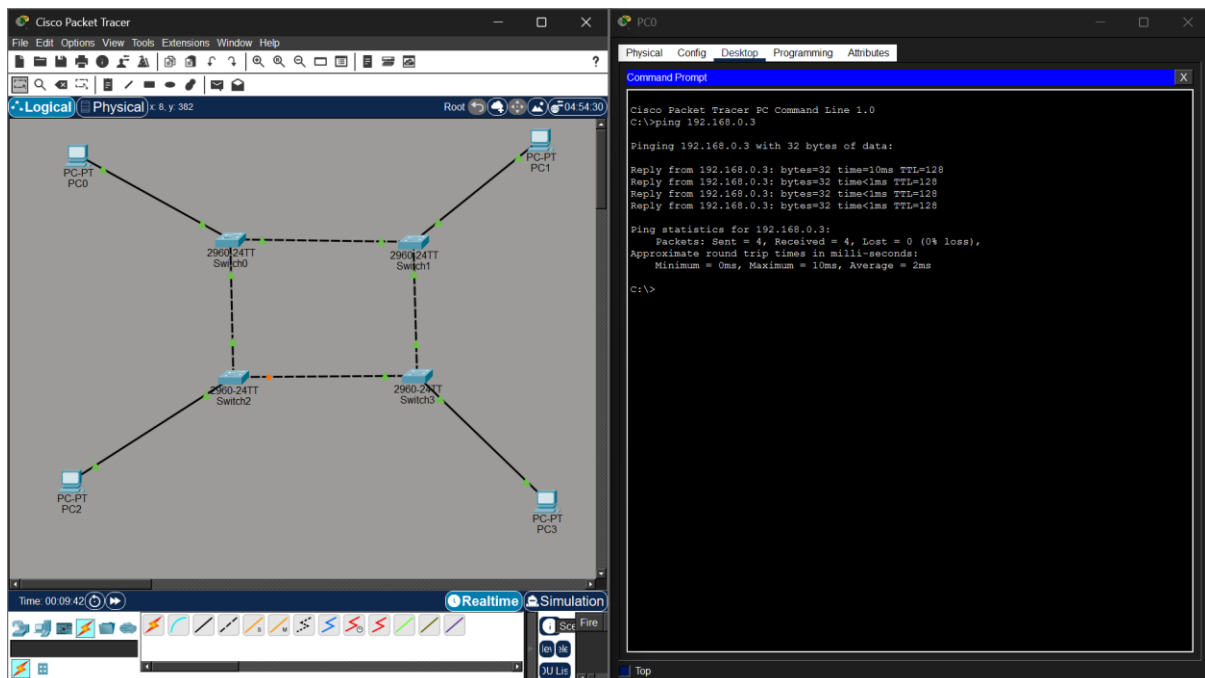
Experiment-02-Star-Topology



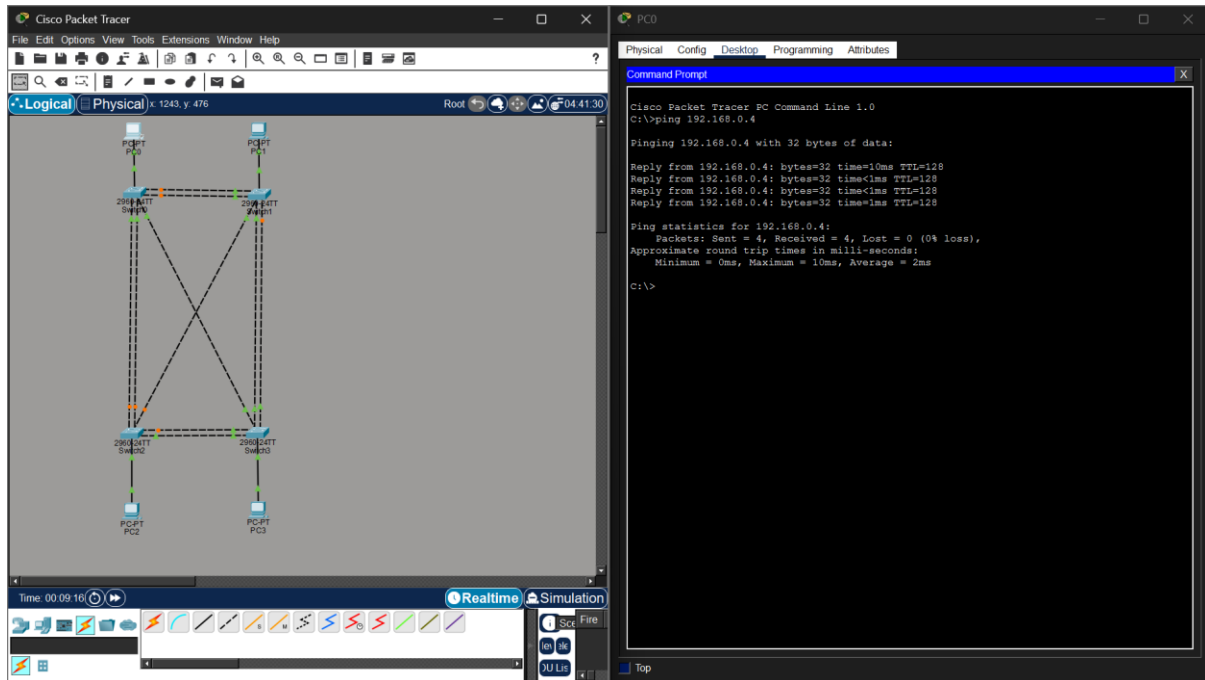
Experiment-03-Bus-Topology



Experiment-04-Ring-Topology



Experiment-05-Mesh-Topology



Experiment-06-Tree-Topology

Cisco Packet Tracer interface showing a tree topology. The central device is a 2503N switch, connected to two 2950 switches. These 2950 switches are connected to six PCs (PC0-PC5). The PDU List Window shows four successful ICMP events:

File	Last Status	Source	Destination	Type	Color	Time/Sec	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC5	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC6	PC0	ICMP		0.000	N	3	(edit)	(delete)

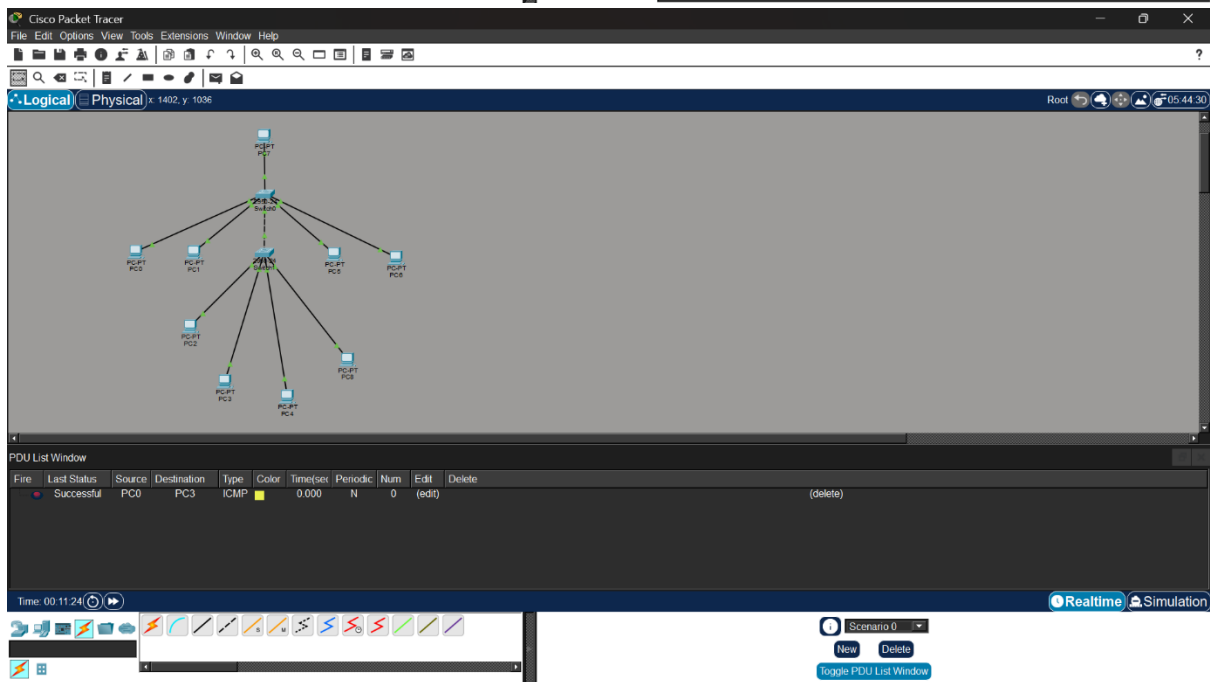
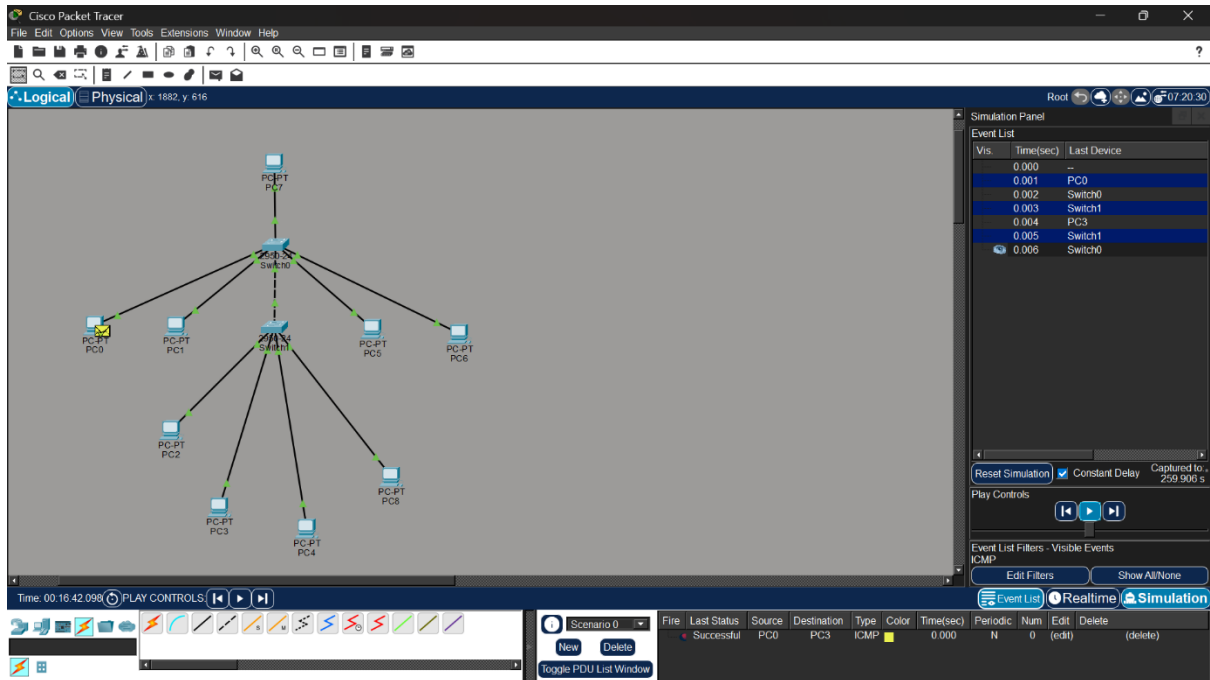
Time: 00:17:34.731

Cisco Packet Tracer interface showing the same tree topology. The PDU List Window shows four successful ICMP events:

File	Last Status	Source	Destination	Type	Color	Time/Sec	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	(delete)
	Successful	PC1	PC4	ICMP		0.000	N	1	(edit)	(delete)
	Successful	PC2	PC5	ICMP		0.000	N	2	(edit)	(delete)
	Successful	PC6	PC0	ICMP		0.000	N	3	(edit)	(delete)

Time: 00:14:57

Experiment-07-Hybrid-Topology



Experiment-08-ARP-Analysis

The image displays two screenshots of the Cisco Packet Tracer interface, illustrating a network setup and the process of analyzing ARP requests.

Top Screenshot:

- Network Topology:** A central Hub0 is connected to six PCs (PC-PT PC0 through PC-PT PC5). PC0 is highlighted in blue.
- PDU List Window:** Shows a single entry for a successful ICMP packet from PC0 to PC3. The table below summarizes this entry:

File	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC0	PC3	ICMP	Orange	0.000	N	0	(edit)	(delete)

Bottom Screenshot:

- Network Topology:** The same network setup is shown, but with green checkmarks on all PC icons, indicating they are powered on.
- Simulation Panel:** The Event List is visible, showing a sequence of events. The table below summarizes the first few events:

Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC0
	0.002	Hub0
	0.002	Hub0
	0.002	Hub0
	0.002	Hub0
	0.003	PC3
	0.004	Hub0
	0.004	Hub0
	0.004	Hub0
	0.004	Hub0

The bottom screenshot also shows the Play Controls and Event List Filters, indicating the simulation is running in Realtime mode.

Experiment-09-CSMA-CD-CSMA-CA

The image displays two screenshots of the Cisco Packet Tracer interface, illustrating the setup and execution of a network experiment.

Top Screenshot: Shows the network topology and the Command Prompt window.

- Network Topology:** A central switch (Switch0) is connected to three PCs (PC0, PC1, PC2). The switch is configured with IP 192.168.1.1 and Subnet Mask 255.255.255.0. The PCs are configured with IP addresses 192.168.1.2, 192.168.1.3, and 192.168.1.4 respectively, all with Subnet Mask 255.255.255.0.
- Command Prompt:** The Command Prompt window shows the configuration of the switch and the execution of ping commands from PC0 to PC1 and PC2.

Bottom Screenshot: Shows the network topology and the Simulation Panel window.

- Network Topology:** The same network topology as the top screenshot.
- Simulation Panel:** The Simulation Panel window shows the event list and the simulation controls. The event list shows the following events:

Vis.	Time(sec)	Last Device
	0.000	-
	0.001	PC0
	0.002	Switch0
	0.003	PC1
	0.004	Switch0

The simulation controls show the simulation running in Realtime mode. The event list filters show the following events:

- ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IEC, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, MODBUS, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, Page, POP3, PPP, PPPoE, PTP, Profile, RADIUS, REP, RFP, RFPv6, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP.

The simulation results show the following events:

Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
PC0	PC1	ICMP		0.000	N	0	(edit)	(delete)

Experiment-10-Computer-Lab

The image displays two screenshots of the Cisco Packet Tracer application. The top screenshot shows a network diagram with a central 2960-24 Switch connected to four PC-PT devices (PC0, PC1, PC2, PC3) and a 1841 Router. The bottom screenshot shows the same network diagram with a Command Prompt window open on PC0, displaying the results of an IPv6 configuration and ping tests.

Network Diagram (Top Screenshot):

- Router: 1841 Router0
- Switch: 2960-24 Switch0
- PCs: PC-PT PC0, PC-PT PC1, PC-PT PC2, PC-PT PC3

Command Prompt Output (Bottom Screenshot):

```
C:\>ipconfig

FastEthernet0 Connection (default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::230:F2FF:FETA:2C39
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 192.168.1.1
    Subnet Mask . . . . .: 255.255.255.0
    Default Gateway . . . . .: ::

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address . . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: 0.0.0.0

C:\>ping 192.168.1.254

Pinging 192.168.1.254 with 32 bytes of data:

Reply from 192.168.1.254: bytes=32 time=12ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255
Reply from 192.168.1.254: bytes=32 time<1ms TTL=255

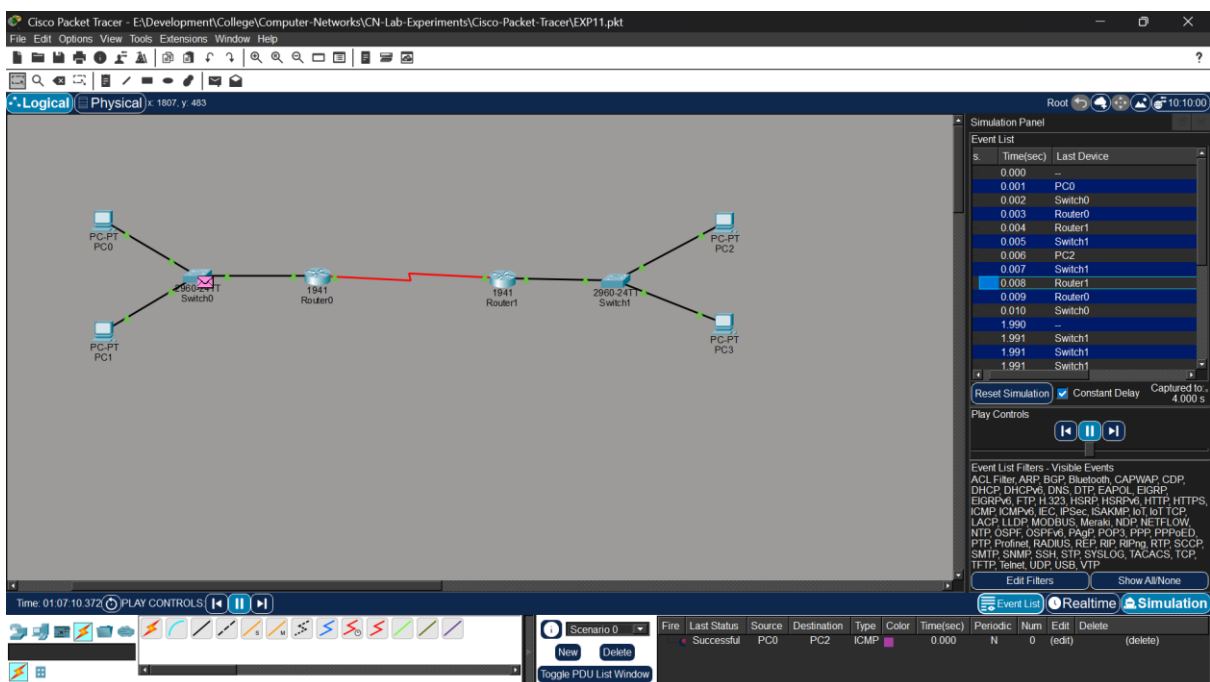
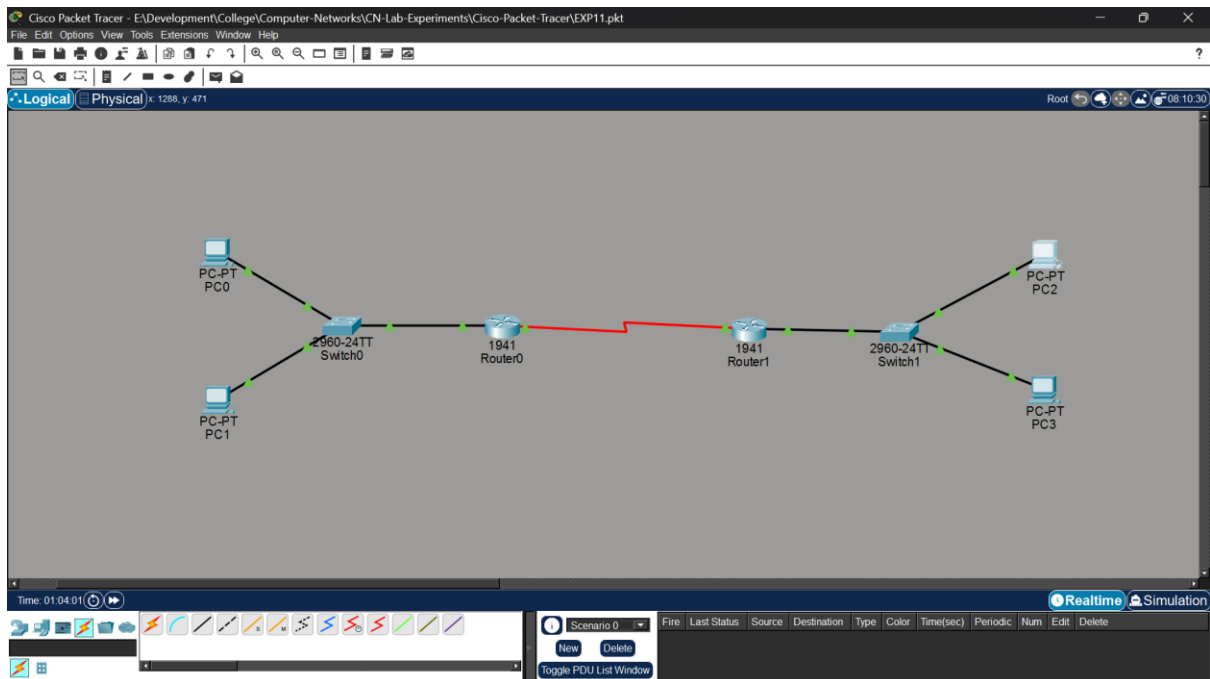
Ping statistics for 192.168.1.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

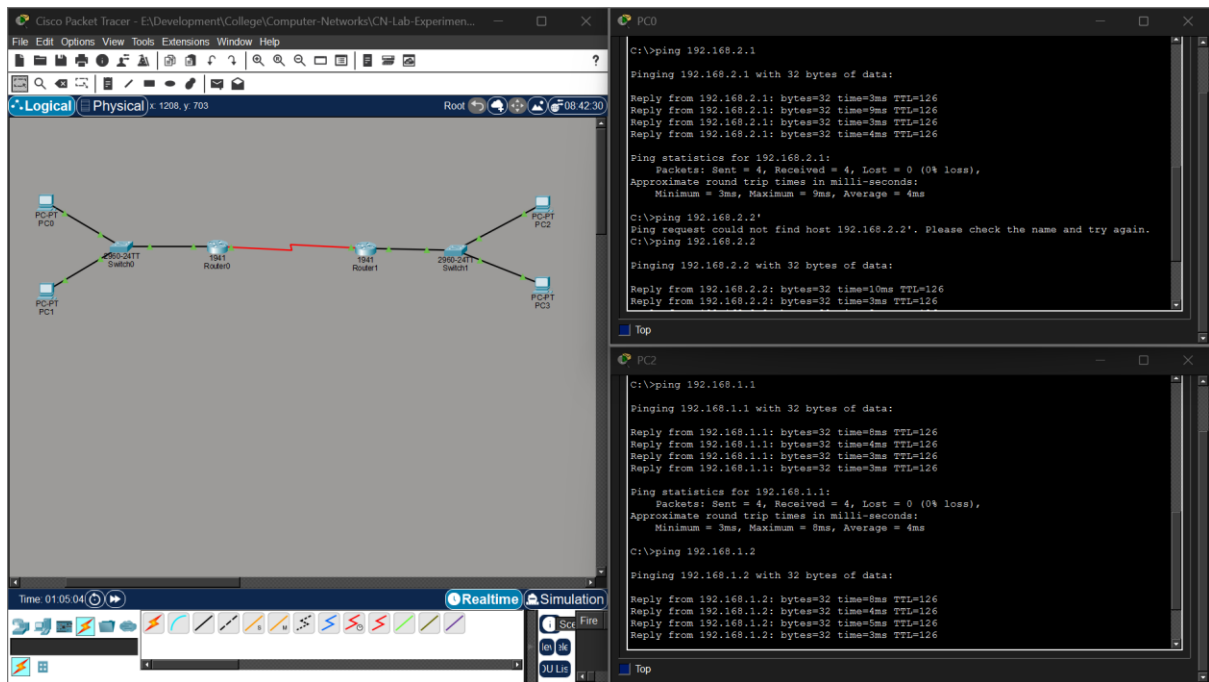
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

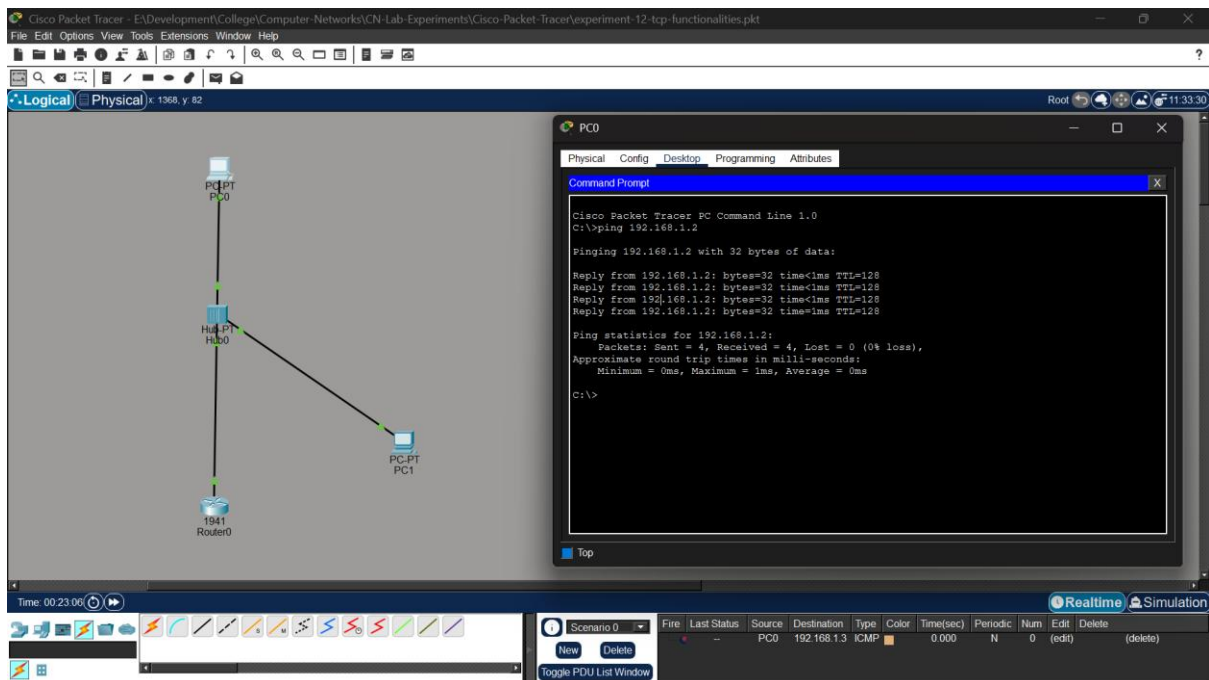
Reply from 192.168.1.4: bytes=32 time=1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
```


Experiment-11-Computer-Lab Router to Router Communication Using Static Routing

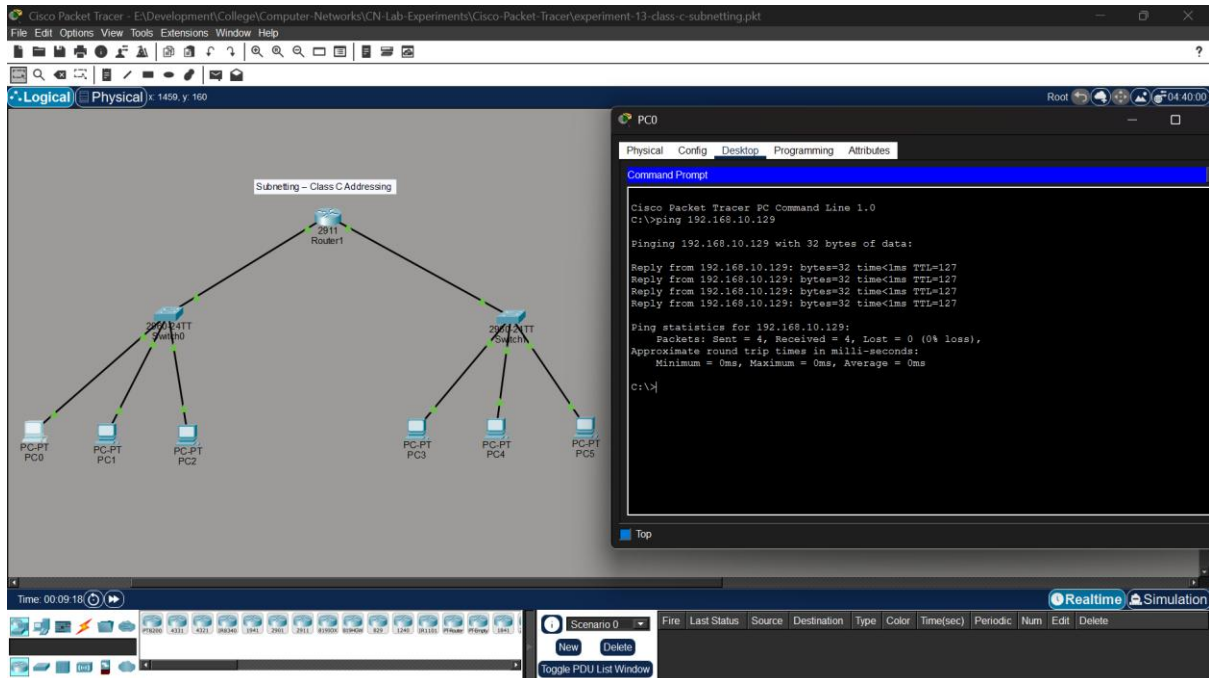




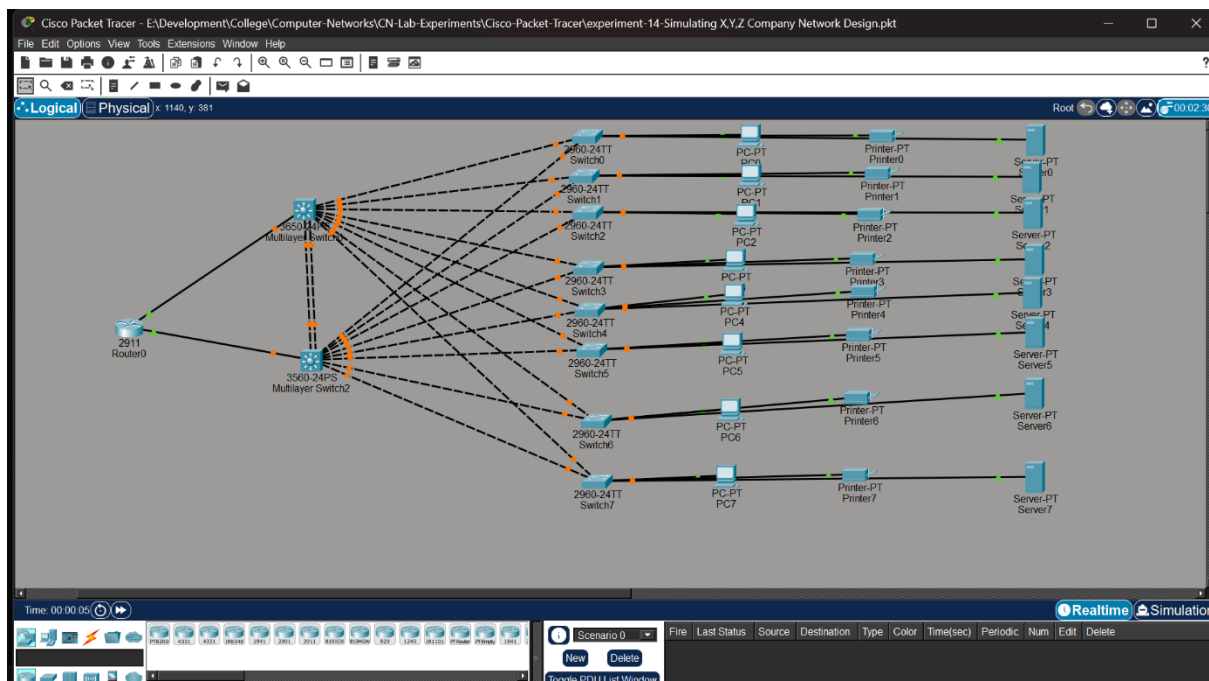
Experiment-12-Tcp-Functionalities



Experiment-13- Experiment-class-c-subnetting



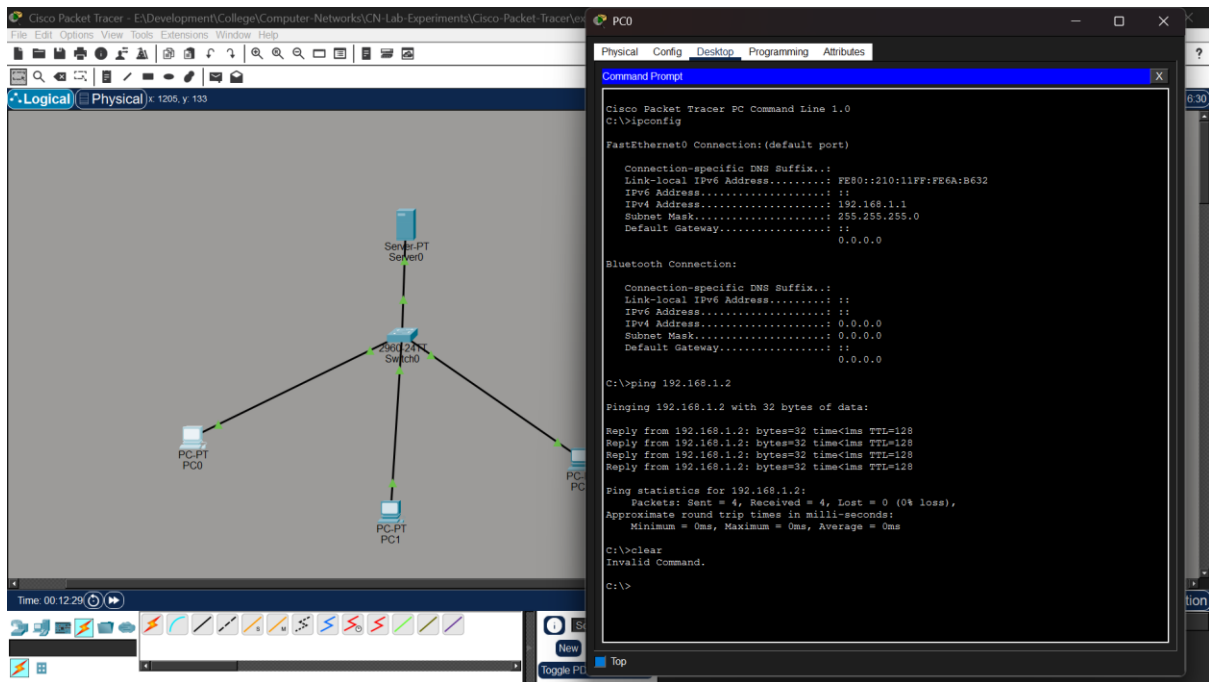
Experiment-14- xyz-company-network-design



Experiment-15- DHCP Configuration

The image displays the Cisco Packet Tracer interface. On the left, a network topology is shown in the 'Physical' tab, featuring a central '2960 24-T Switch' connected to a 'Server-PT Server0' and three 'PC-PT' devices (PC0, PC1, PC2). The 'Logical' tab is also visible. On the right, the 'PC0' configuration window is open, showing the 'Config' tab. The 'IP Configuration' section is active, with 'DHCP' selected. The 'IPv4 Address' is set to '192.168.1.1', the 'Subnet Mask' is '255.255.255.0', and the 'Default Gateway' is '0.0.0.0'. The 'DNS Server' is also set to '0.0.0.0'. The 'IPv6 Configuration' section is also visible, with 'Automatic' selected. The '802.1X' section is expanded, showing 'Use 802.1X Security' checked, 'Authentication' set to 'MD5', and fields for 'Username' and 'Password'.

The image shows the same Cisco Packet Tracer interface as above, but with the simulation running. The network topology remains the same. The 'Realtime' button is highlighted in the bottom right corner, indicating that the simulation is active. The 'Simulation' button is also visible. The 'Scenario 0' dropdown menu is open, showing 'New' and 'Delete' options. The 'Toggle PDU List Window' button is also visible.



Experiment-15- DHCP Configuration

